CHAPTER 25

# The Urinary System

How the kidneys filter the blood and make urine.

Anatomy & Physiology · Broward-Miami Health Institute

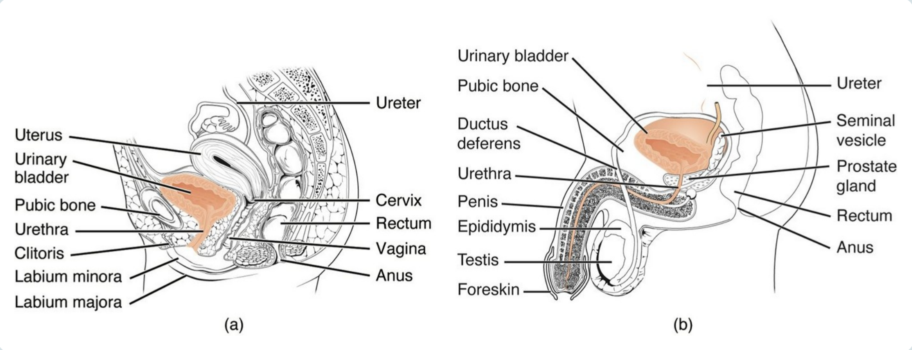
FACULTY EDITION

CHAPTER 25

# Learning Objectives

*After studying this chapter, students will be able to:*

1. Describe the composition of urine
2. Label structures of the urinary system
3. Characterize the roles of each of the parts of the urinary system
4. Illustrate the macroscopic and microscopic structures of the kidney
5. Trace the flow of blood through the kidney
6. Outline how blood is filtered in the kidney nephron




*The organs of the urinary system.*

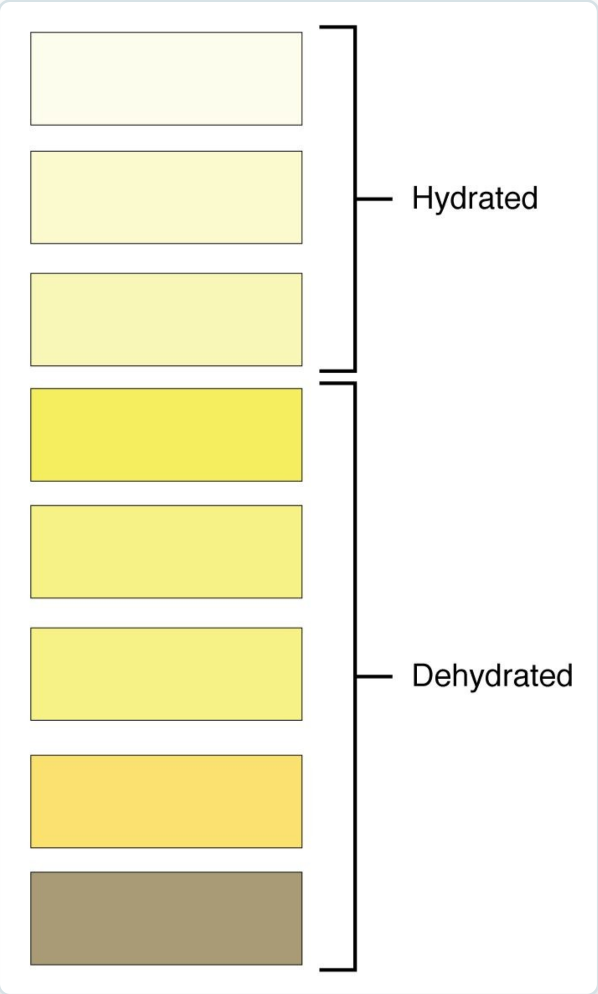
25.1

SECTION

# Physical Characteristics of Urine

What urine reveals about the body's internal balance.

# Characteristics of Urine




*Urine color and hydration*

- Urine reflects the body's state
- Color signals hydration
- Normally clear and yellow
- Tested in urinalysis

25.2

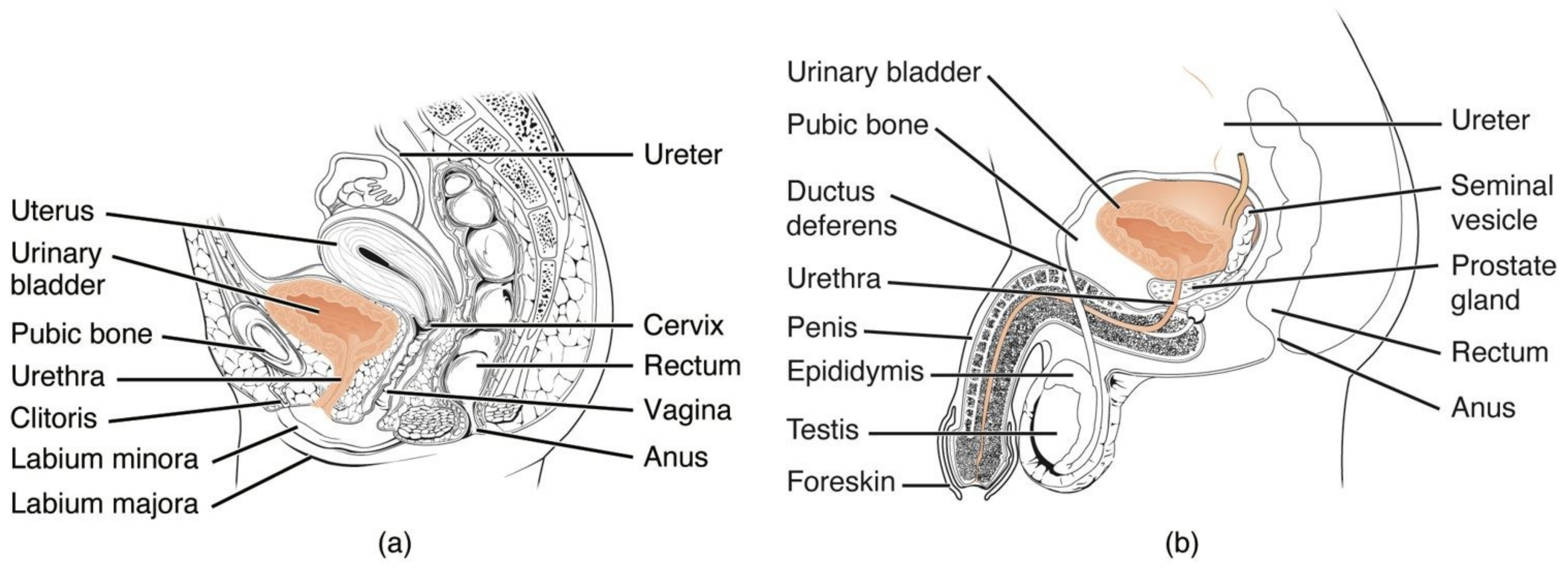
SECTION

# Gross Anatomy of Urine Transport

The ureters, bladder, and urethra that store and transport urine.

SECTION 25.2

# The Urinary System




*The organs of the urinary system*

- Two kidneys make urine
- Ureters carry it to the bladder
- The bladder stores it
- The urethra empties it out

25.3

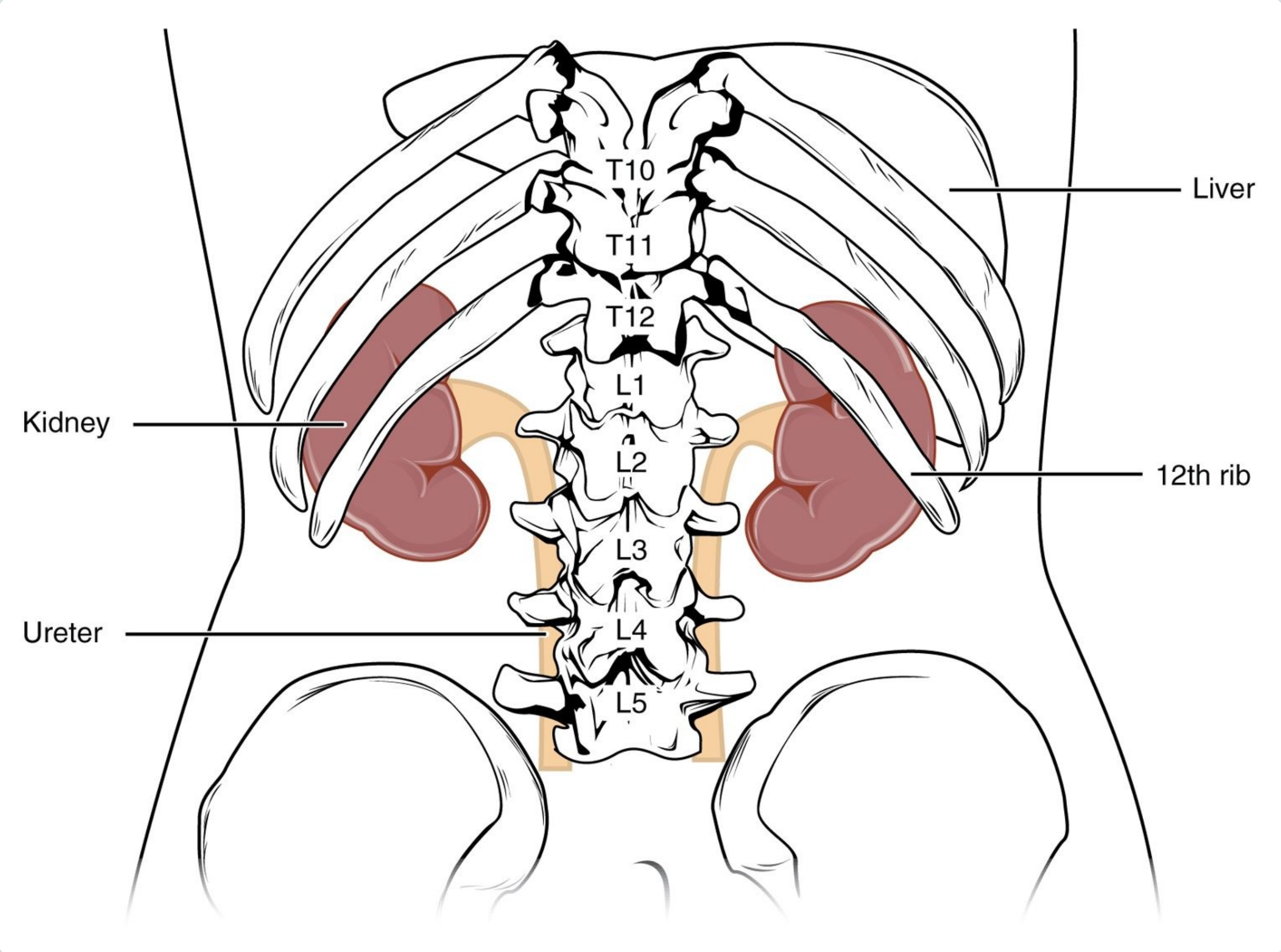
SECTION

# Gross Anatomy of the Kidney

The position and internal structure of the kidneys.

# Position of the Kidneys

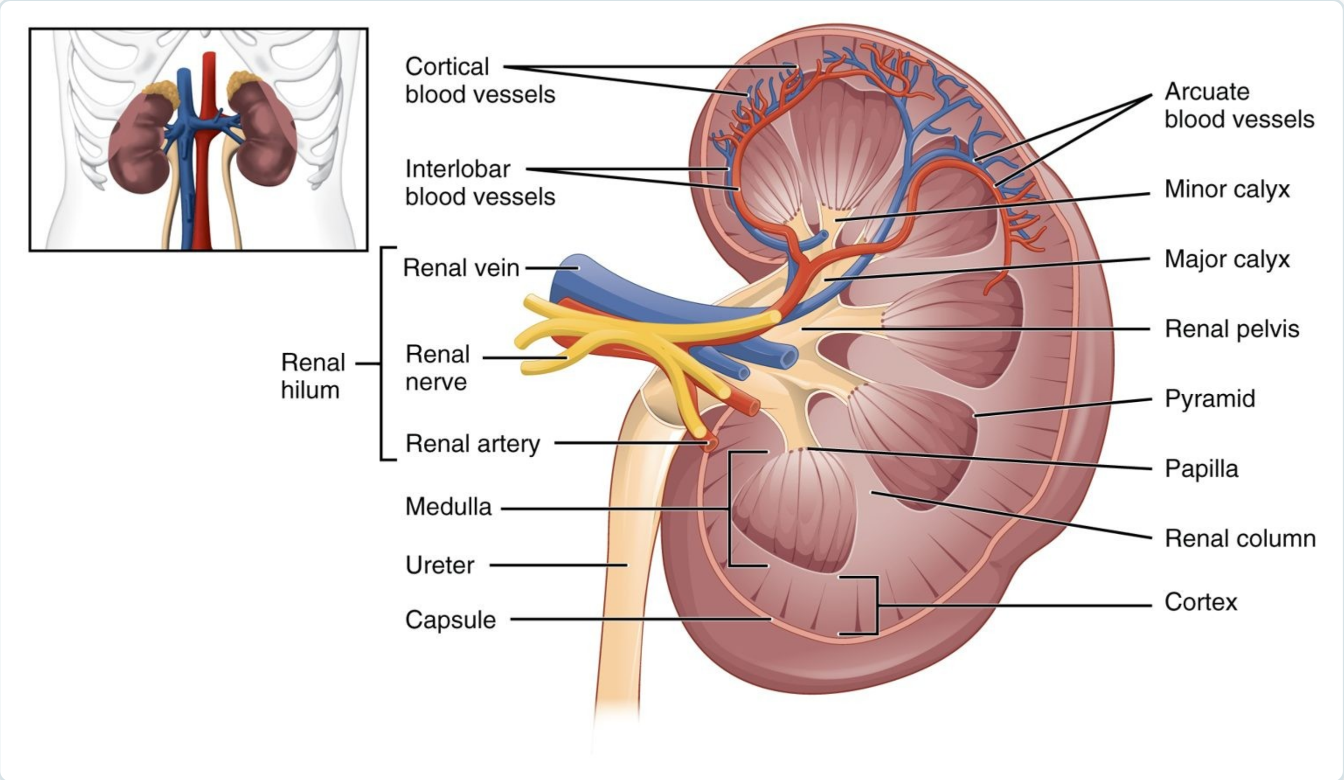
- Two bean-shaped organs
- Sit against the back wall
- Retroperitoneal (behind the abdomen)
- Partly protected by the lower ribs




*The position of the kidneys*

SECTION 25.3

# Internal Kidney Anatomy




*Internal anatomy of the kidney*

- Outer cortex, inner medulla
- Medulla forms renal pyramids
- Urine collects in the renal pelvis
- Drains into the ureter

25.4

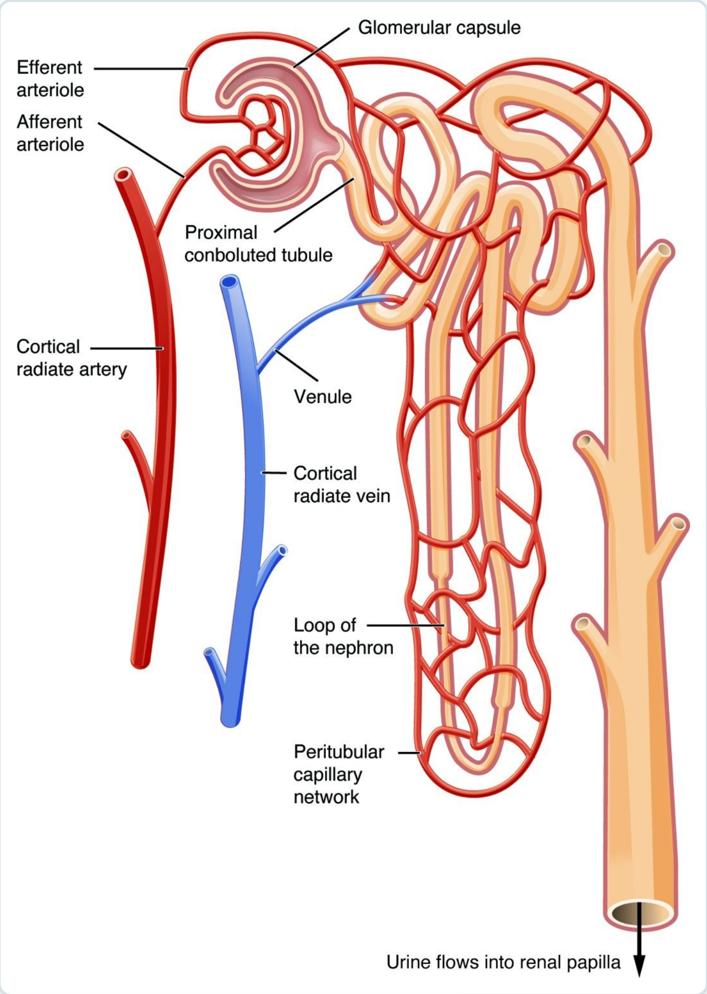
SECTION

# Microscopic Anatomy of the Kidney

The nephron — the kidney's microscopic filtering unit.

SECTION 25.4

# The Nephron




*The structure of a nephron*

- The functional unit of the kidney
- About a million per kidney
- Has a glomerulus and a tubule
- Filters blood and forms urine

25.5

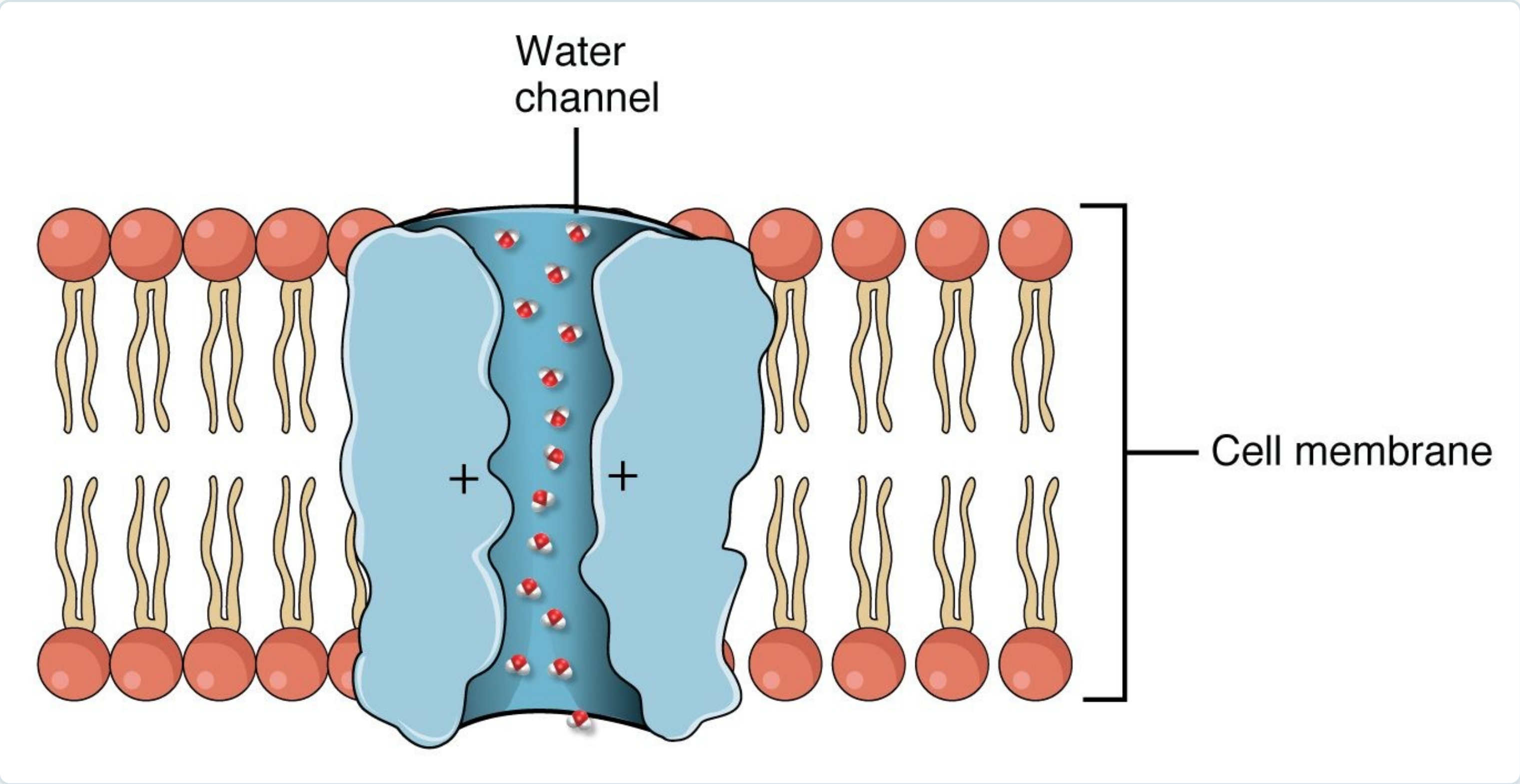
SECTION

# Physiology of Urine Formation

Filtration, reabsorption, and secretion — how urine is made.

SECTION 25.5

# Glomerular Filtration




*Glomerular filtration*

- Urine formation starts with filtration
- Blood pressure pushes fluid out
- Filters water and small molecules
- Keeps cells and proteins in the blood

25.6

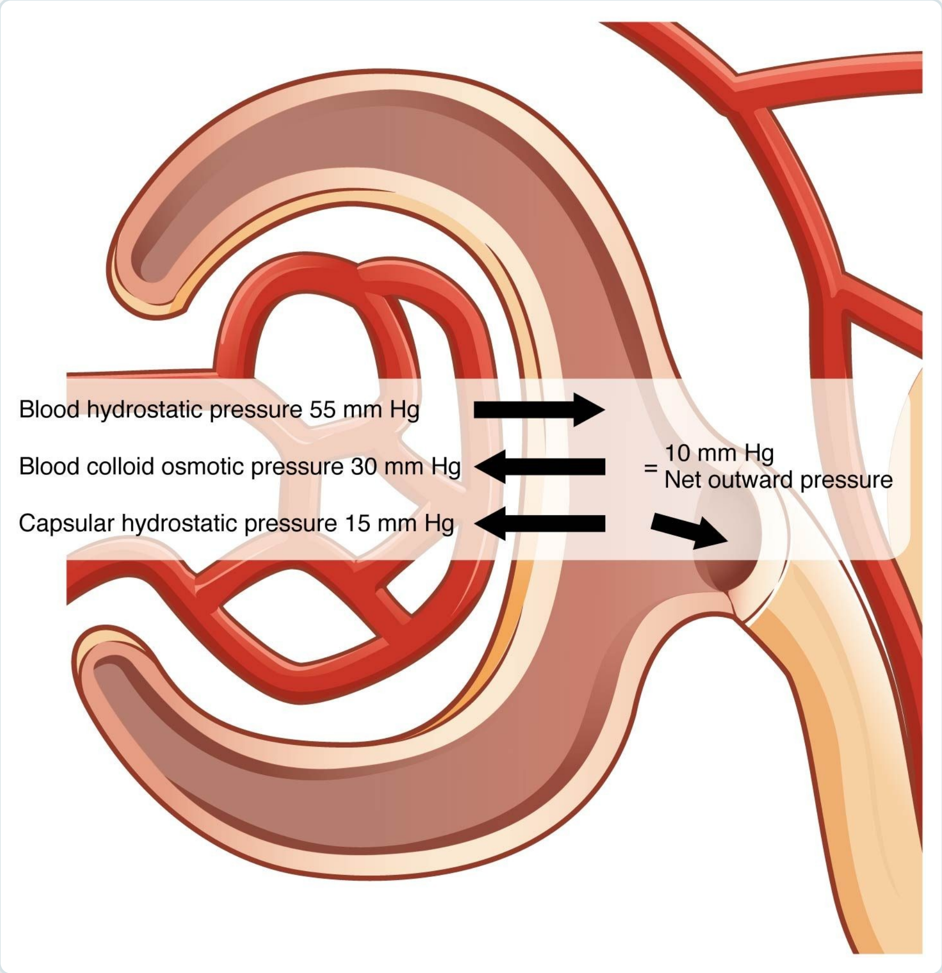
SECTION

# Tubular Reabsorption

Reclaiming what the body needs from the filtrate.

# Reabsorption & Secretion

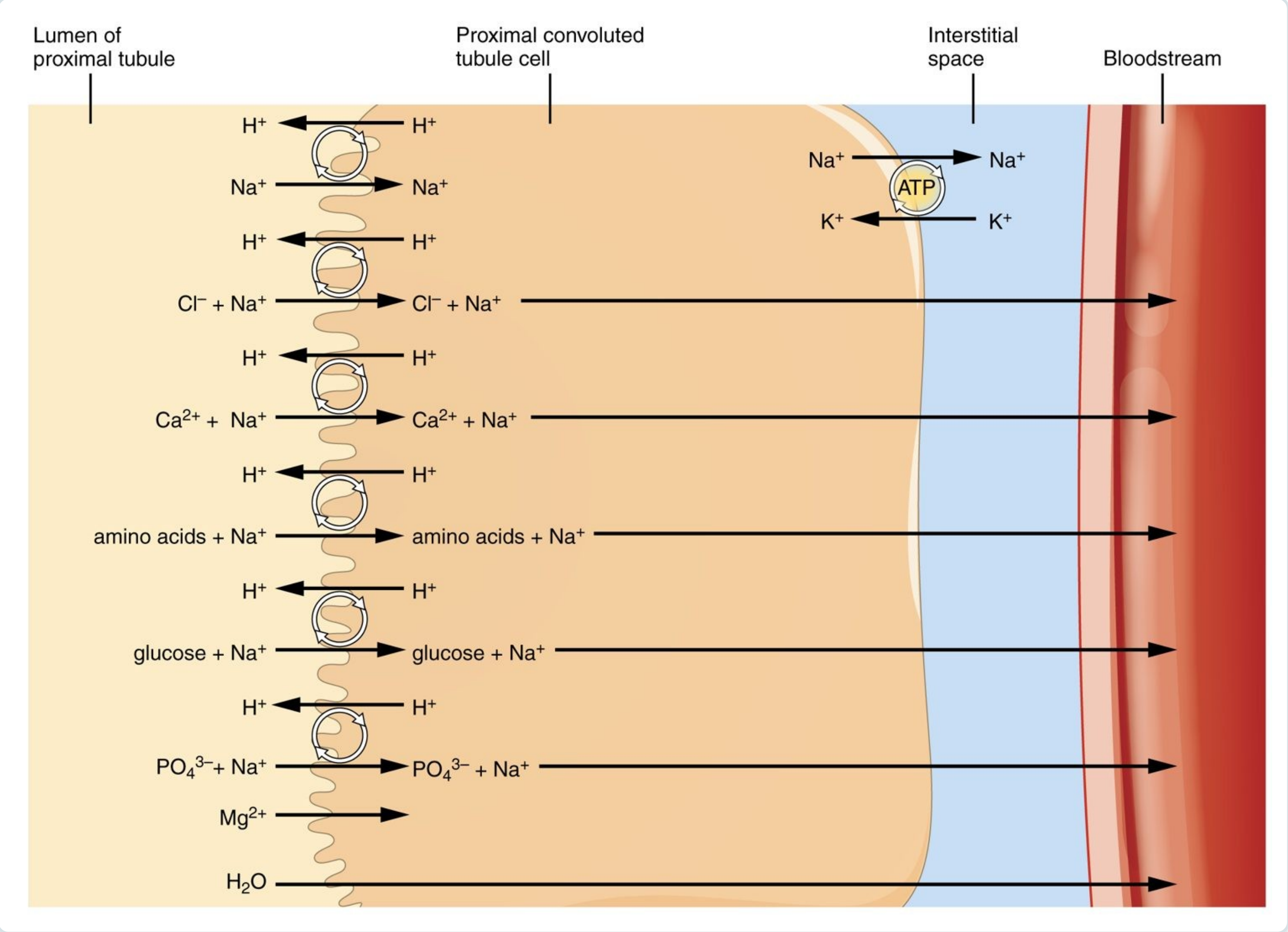
- Most of the filtrate is reclaimed
- Water, glucose, ions are reabsorbed
- Wastes are secreted into the tubule
- Only urine remains




*Reabsorption and secretion along the nephron*

# Concentrating the Urine

- The loop of Henle builds a salt gradient
- The medulla becomes very salty
- Water is drawn out of the tubule
- Produces concentrated urine




*Concentrating the urine in the loop of Henle*

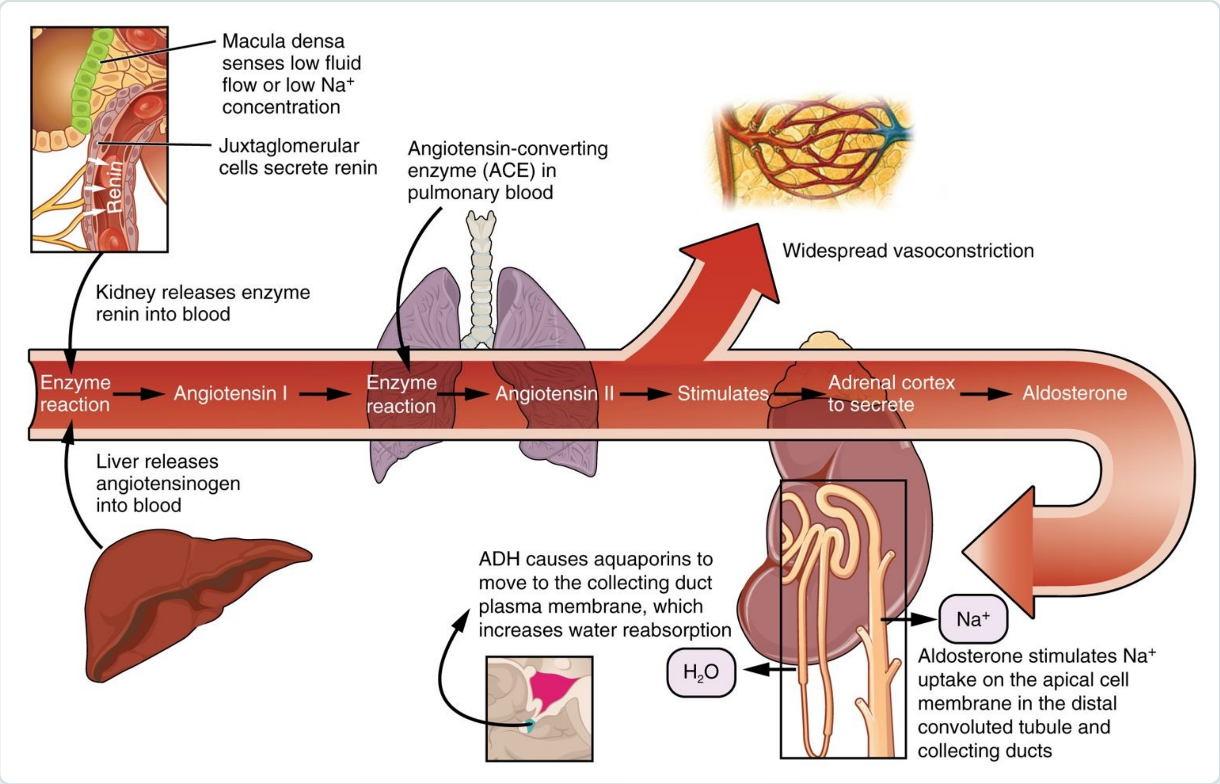
SECTION

# 25.8 Endocrine Regulation of Kidney Function

How hormones fine-tune kidney function and fluid balance.

SECTION 25.8

# Hormonal Regulation




*Endocrine regulation of the kidney*

- Hormones adjust water and salt
- ADH increases water reabsorption
- Aldosterone retains sodium
- Together they control blood volume and pressure

# Key Takeaways

- The kidneys filter blood and make urine; the ureters, bladder, and urethra store and remove it.
- Each kidney has a cortex and medulla and about a million nephrons.
- Urine forms by filtration, reabsorption, and secretion along the nephron.
- The loop of Henle lets the kidney concentrate urine and conserve water.
- Hormones like ADH and aldosterone adjust fluid and electrolyte balance.

CLINICAL CONNECTIONS

# Why the Urinary System Matters in Practical Nursing

### Urinary Tract Infection

Common, especially in women — watch for symptoms.

### Kidney Stones

Hardened deposits cause severe flank pain.

### Kidney Failure

Lost filtration causes waste buildup; may need dialysis.

### Urinalysis

A simple test revealing many systemic conditions.

### Fluid Balance

The kidneys are central to blood volume and pressure.

### Diuretics

Drugs that increase urine output to manage fluid overload.